① $(1+e^{x/y})dx + e^{\frac{x}{y}}(1-\frac{x}{y})dy = 0$

Soln: Given differential Equation,

$$(1+e^{\frac{x}{y}})dx + e^{\frac{x}{y}}(1-\frac{x}{y})dy = 0 \longrightarrow (1)$$

which is in the form

$$M(x,y)dx + N(x,y)dy = 0$$

where,

$$M = 1+e^{\frac{x}{y}}$$

$$N = e^{\frac{x}{y}}(1-\frac{x}{y})$$

$$\frac{\partial M}{\partial y} = e^{\frac{x}{y}} \times \left(-\frac{1}{y^2}\right)$$

$$\frac{\partial M}{\partial y} = e^{\frac{x}{y}} \left(-\frac{x}{y^2}\right)$$

$$\therefore \frac{\partial M}{\partial y} = -\frac{x}{y^2} e^{\frac{x}{y}}$$

$$\frac{\partial N}{\partial x} = e^{\frac{x}{y}} - x e^{\frac{x}{y}} \left(\frac{1}{y}\right)$$

$$+ \left[\left(\frac{1}{y}\right) y e^{\frac{x}{y}} \frac{1}{x}\right] - \left[\frac{1}{x} \cdot e^{\frac{x}{y}} \cdot 1\right]$$

$$\frac{\partial N}{\partial x} = \frac{1}{y} e^{\frac{x}{y}} - \frac{1}{y} e^{\frac{x}{y}} - \frac{x}{y^2} e^{\frac{x}{y}}$$

$$\therefore \frac{\partial N}{\partial x} = -\frac{x}{y^2} e^{\frac{x}{y}}$$

$\therefore$ (1) is an Exact D.E

General Solution is,

$$\int_{y \text{ constant}} M(x,y)\, dx + \int (\text{Terms not containing } x \text{ in } N)\, dy = C$$

$$\int_{y \text{ constant}} \left(1 + e^{\frac{x}{y}}\right) dx + \int 0 \, dy = C$$

$$\left(\int 1 + \int e^{\frac{x}{y}}\right) dx = C$$

$$x + \frac{e^{\frac{x}{y}}}{\left(\frac{1}{y}\right)} = C$$

$$\therefore x + y e^{\frac{x}{y}} = C$$

② $(1+2xy\cos x^2-2xy)\,dx + (\sin x^2 - x^2)\,dy = 0$.

Sol: Given differential Equation,

$(1+2xy\cos x^2-2xy)dx + (\sin x^2-x^2)dy=0$ ①

which is on the form

$M(x,y)\,dx + N(x,y)\,dy = 0$

where,

$M = 1+2xy\cos x^2 - 2xy$

$N = \sin x^2 - x^2$

$\frac{\partial M}{\partial y} = 0 + 2x\cos x^2 \cdot 1 - 2x \cdot 1$

$\therefore \frac{\partial M}{\partial y} = 2x\cos x^2 - 2x$

$\frac{\partial N}{\partial x} = \cos x^2 (2x) - 2x$

$\therefore \frac{\partial N}{\partial x} = 2x\cos x^2 - 2x$

$\therefore$ ① is an Exact DE.

General solution is given by

$\int_{y\text{ constant}} M(x,y)\,dx + \int_{y\text{ constant}} (\text{Terms not containing } x \text{ in } N)\,dy = C$

$\int (1+2xy\cos x^2 - 2xy)\,dx + \int 0\,dy = C$

$\int dx + \int 2xy\cos x^2\,dx - \int 2xy\,dx = C$

$x + y\int \cos t\,dt - 2y\cdot\frac{x^2}{2} = C$

$x + y\sin t - x^2 y = C.$

$\therefore x + y\sin x^2 - x^2 y = C.$

$x^2 = t$

$2x\,dx = dt$

③ $(y^2 e^{xy^2} + 4x^3)\,dx + (2xy\,e^{xy^2} - 3y^2)\,dy = 0$

Sol:- Given differential Equation,

$(y^2 e^{xy^2} + 4x^3)\,dx + (2xy\,e^{xy^2} - 3y^2)\,dy = 0 \rightarrow (1)$

which is in the form

$M(x,y)\,dx + N(x,y)\,dy = 0$

where,

$M = y^2 e^{xy^2} + 4x^3$

$N = 2xy\,e^{xy^2} - 3y^2$

$\frac{\partial M}{\partial y} = (2y)e^{xy^2} + (y^2)(2xy\,e^{xy^2})$

$= 2y\,e^{xy^2} + 2xy^3 e^{xy^2}$

$= 2y\,e^{xy^2}(1 + xy^2)$

$\frac{dM}{dy} = 2y\,e^{xy^2}(1 + xy^2)$

$\frac{dN}{dx} = 2y\,e^{xy^2} + 2xy^3 e^{xy^2}$

$= 2y\,e^{xy^2}(1 + xy^2)$

$\therefore$ (1) is an exact D.E

General Solution is,

$\int_{y\text{constant}} M(x,y)\,dx + \int (\text{Terms not containing } x \text{ in } N)\,dy = C$

$\int_{\text{const}} y^2 e^{xy^2} + 4x^3\,dx + \int (-3y^2)\,dy$

$\int y^2 e^{xy^2}\,dx + \int 4x^3\,dx + \int -3y^2\,dy$

$\because \boxed{\int e^{f(x)} f'(x)\,dx = e^{f(x)}}$

$e^{xy^2} + 4\frac{x^4}{4} - 3\frac{y^3}{3}$

$\therefore e^{xy^2} + x^4 - y^3 + C.$

(4) $(xe^{xy}+2y)\frac{dy}{dx}+ye^{xy}=0$

Sol:- Given differential Equation,

$(xe^{xy}+2y)\frac{dy}{dx}+ye^{xy}=0$ —

$(xe^{xy}+2y)\frac{dy}{dx}=-ye^{xy}$

$(xe^{xy}+2y)dy=-ye^{xy}dx$

$(ye^{xy})dx+(xe^{xy}+2y)dy=0 \rightarrow (1)$

which is in the form

$M(x,y)dx+N(x,y)dy=0$

Where,

$M=ye^{xy}$

$N=xe^{xy}+2y$

$\frac{dM}{dy}=(1)e^{xy}+ye^{xy}(x)$

$=e^{xy}(1+xy)$

$\frac{dN}{dx}=1e^{xy}+xe^{xy}(y)$

$=e^{xy}(1+xy)$

$\therefore$ (1) is an Exact D.E.

General Solution is,

$\int_{y\,\text{const}} M(x,y)\,dx+\int(\text{Terms not containing } x \text{ in } N)\,dy=c$

$\int ye^{xy}\,dx+\int 2y\,dy$

$\because \boxed{\int e^{f(x)}f'(x)\,dx=e^{f(x)}}$

$e^{xy}+\frac{xy^2}{x}+C$

$\therefore e^{xy}+y^2+C.$

(5)

Sol:-

(5) $\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$

Sol:- Given differential equation,

$$\frac{dy}{dx} + \frac{y\cos x + \sin y + y}{\sin x + x\cos y + x} = 0$$

$$\Rightarrow (y\cos x + \sin y + y)\frac{dy}{dx} = -(y\cos x + \sin y + y)$$

$$(\sin x + x\cos y + x)dy = -(y\cos x + \sin y + y)\,dx$$

$$(\sin y + y\cos x + y)dx + (\sin x + x\cos y + x)dy = 0 \rightarrow (1)$$

which is in the form

$M(x,y)\,dx + N(x,y)\,dy = 0$

where,

$M = \sin y + y\cos x + y$

$N = \sin x + x\cos y + x$

$\frac{dM}{dy} = \cos y + (\cos x \cdot 1 + 1)$

$= \cos y + \cos x + 1$

$\frac{dN}{dx} = \cos x + 1\cdot(\cos y + 1)$

$= \cos x + \cos y + 1$

$\therefore$ (1) is an exact D.E

General solution is,

$$\int_{y\,\text{const}} M(x,y)\,dx + \int \left(\begin{matrix}\text{Terms not}\\ \text{containing}\\ x \text{ in } N\end{matrix}\right) dy = C$$

$\int \sin y + y\cos x + y \; dx$

$\therefore x\sin y + y\sin x + xy + C$

(6) $[y(1+\frac{1}{x})+\cos y]dx + (x+\log x - x\sin y)dy = 0$

Sol: Given differential Equation,

$$\left[y\left(1+\frac{1}{x}\right)+\cos y\right]dx + \left(x+\log x - x\sin y\right)dy = 0 \longrightarrow (1)$$

Which is in the form

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

Where,

$$M = y\left(1+\frac{1}{x}\right)+\cos y$$
$$= y + \frac{y}{x} + \cos y$$
$$N = x + \log x - x\sin y$$

$$\frac{dM}{dy} = 1 + \frac{1}{x} - \sin y$$

$$\frac{dN}{dx} = 1 + \frac{1}{x} - \sin y$$

$\therefore$ (1) is an Exact DE

General solution is,

$$\int_{y\text{ const}} M(x,y)\,dx + \int \left(\begin{array}{c}\text{Terms not}\\ \text{Containing}\\ x \text{ in } N\end{array}\right) dy = c$$

$$\int y\left(1+\frac{1}{x}\right)dx + \int 0\,dy = c$$

$$yx + y\log x + C$$

$\therefore xy + y\log x + C$

(7) $\frac{2x}{y^3}\,dx + \left(\frac{y^2-3x^2}{y^4}\right)dy = 0$

Sol: Given differential Equation,

$$\frac{2x}{y^3}\,dx + \left(\frac{y^2-3x^2}{y^4}\right)dy = 0 \rightarrow (1)$$

which is in the form

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

where,

$$M = \frac{2x}{y^3} = 2xy^{-3}$$

$$N = \frac{y^2-3x^2}{y^4}$$

$$\frac{dM}{dy} = -6xy^{-4}$$

$$= \frac{-6x}{y^4}$$

$$\frac{dN}{dx} = \frac{-6x}{y^4}$$

$\therefore$ (1) is an Exact D.E

General solution is,

$$\int_{y\,\text{const}} M(x,y)\,dx + \int \left(\begin{array}{c}\text{Terms not}\\ \text{Containing}\\ x \text{ in } N\end{array}\right) dy = 0$$

$$\int \frac{2x}{y^3}\,dx + \int \frac{1}{y^2}\,dy$$

$$\frac{2}{y^3}\int x + \int \frac{1}{y^2}\,dy$$

$$\frac{2}{y^3}\,\frac{x^2}{2} + \int y^{-2}\,dy$$

$$\frac{x^2}{y^3} + \frac{y^{-1}}{1} + C$$

$$\therefore \frac{x^2}{y^3} - \frac{1}{y} + C$$

8) $x^3 \sec^2 y \frac{dy}{dx} + 3x^2 \tan y = \cos x$

Sol:- Given differential Equation,

$$x^3 \sec^2 y \frac{dy}{dx} + 3x^2 \tan y = \cos x$$

$$(x^3 \sec^2 y)\frac{dy}{dx} = \cos x - 3x^2 \tan y$$

$$(x^3 \sec^2 y)\,dy = (\cos x - 3x^2 \tan y)\,dx$$

$$(3x^2 \tan y - \cos x)\,dx + (x^3 \sec^2 y)\,dy = 0 \quad \text{---} (1)$$

which is in the form

$$M(x,y)\,dx + N(x,y)\,dy = 0$$

where,

$M = 3x^2 \tan y - \cos x$

$N = x^3 \sec^2 y$

$\frac{dM}{dy} = 3x^2 \sec^2 y$

$\frac{dN}{dy} = 3x^2 \sec^2 y$

$\therefore$ (1) is an Exact DE

General solution is,

$$\int_{y\text{ constant}} M(x,y)\,dx + \int (\text{Terms not containing } x \text{ in } N)\,dy = 0$$

$$\int 3x^2 \tan y - \cos x \, dx$$

$$\int 3x^2 \tan y \, dx - \int \cos x \, dx$$

$$\frac{3x^3}{3} \tan y - \sin x + C$$

$\therefore x^3 \tan y - \sin x + C$